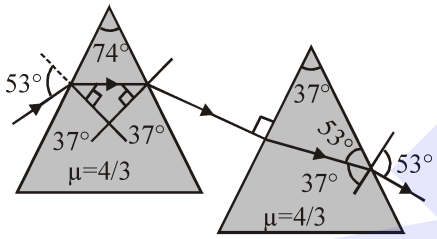
SOLUTION
SECTION-I

Single Correct Answer Type **4 Q. [3 M (–1)]**

1. Ans. (A)

Sol. Here we have drawn the ray diagram with all angles. The values of all angles can be found by using concepts of prism theory, refraction and basic geometry.



Deviation produced by 1st prism is, $\delta_1 = 53° + 53° - 74° = 32°$ clockwise
Deviation produced by 2nd prism is, $\delta_2 = 53° - 37° = 16°$ – clockwise
So, $\delta = 32° + 16° = 48°$ clockwise

2. Ans. (A)

Sol. $\frac{1}{v} - \frac{1}{-15} = \frac{1}{5}$; $\frac{1}{v} - \frac{1}{5} - \frac{1}{15}$; $v = 7.5$; $v' = -12.5$

$$\frac{1}{v'} - \frac{1}{12.5} = \frac{1}{-15}; \frac{1}{v'} = \frac{2}{25} - \frac{1}{15}; \frac{1}{v'} = \frac{12-10}{150}$$

$v' = 75$ cm; $v'' = -95$ cm; $\frac{1}{v''} + \frac{1}{95} = \frac{1}{5}$; $v'' = 5.3$ cm

3. Ans. (A)

Sol. $\frac{1}{f_{eq}} = \frac{1}{f_m} - \frac{2}{f_L}$

$$\frac{1}{f_L} = (\mu - 1)\left[\frac{+1}{R}\right] = \frac{1}{2R}$$

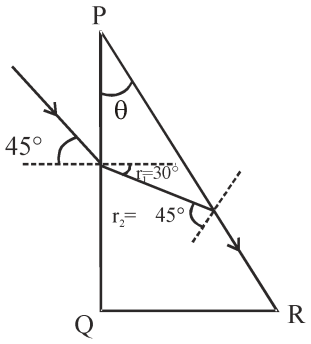
$\therefore \quad \frac{1}{f_q} = -\frac{1}{R} \Rightarrow f_{eq} = -R$ [concave mirror]

$\therefore$ Image of object kept at 2R will form at 2R

4. Ans. (A)

Sol. This case of prism is unlike to the cases we are familiar with. In this case the light ray is coming from above, so our conventional formulas would not be applicable. In the ray diagram below we can say that:
$\Rightarrow (r_1 + 90°) + (90 - r_2) + \theta = 180°$
$\Rightarrow r_2 - r_1 = \theta$. Rest of the solution is as below.

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$1 \sin 45° = \sqrt{2} \sin r_1$

$r_2 - r_1 = \theta$

$\theta = 45° - 30°$

$\Rightarrow \theta = 15°$

Multiple Correct Answer Type 11 Q. [4 M (–1)]

5. Ans. (A,B,C,D)

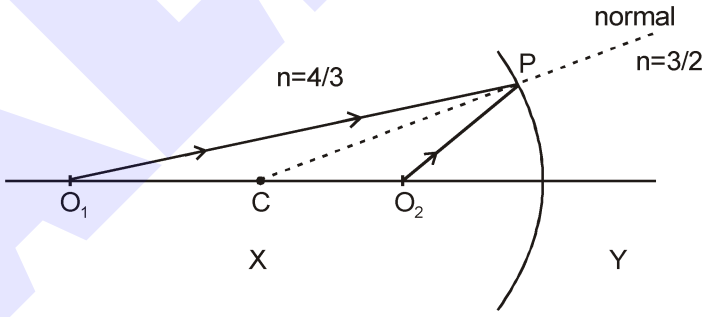
Sol. $\delta_{min} = 2i - A \Rightarrow 60° = 2(60°) - A \Rightarrow A = 60°$

$$\mu = \frac{\sin\left(\frac{A+\delta_{min}}{2}\right)}{\sin\left(\frac{A}{2}\right)} = \frac{\sin\left(\frac{60°+60°}{2}\right)}{\sin\left(\frac{60°}{2}\right)} = \sqrt{3} \Rightarrow \delta = i + e - A$$

$65° = i_1 + 70° - 60° \Rightarrow i_1 = 55°$

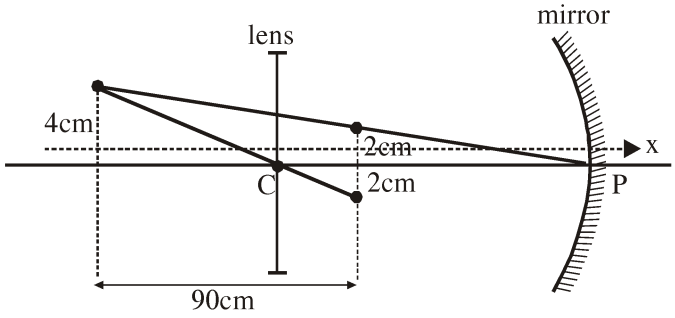
6. Ans. (A,B,D)

Sol. Let there be two point objects O_1 and O_2, Incident rays from O_1 and O_2 at point P shall both bend towards normal and hence the corresponding refracted rays shall intersect the principal axis in the left medium. Therefore image formed under given condition shall always be virtual.



7. Ans. (A,B)

Sol.



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For mirror $\frac{1}{f}=-\frac{1}{90}-\frac{1}{180}=-\frac{1}{60}$ For lens $\frac{1}{f}=\frac{1}{30}+\frac{1}{30}+\frac{1}{60}=\frac{1}{20}$

8. Ans. (A, B, C, D)

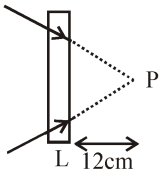
Sol. $\frac{m_1}{m_2}=4\Rightarrow m_1m_2=1\Rightarrow m_1=2, m_2=\frac{1}{2}$

x + y = 96; $\frac{y}{x}$ =2, y = 2x, 3x = 96, x = 32

distance between two positive = x – y = 32

9. Ans. (C, D)

Sol.



(A) If L is convex & focal length 20

f = – 20 u = +12 ⇒ v = +7.5

(C) If L is concave lense

than f = +16 u = +12 ⇒ v = +48

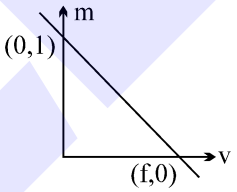
10. Ans. (A,B,C)

Sol. $\frac{1}{v}-\frac{1}{u}=\frac{1}{f}\Rightarrow 1-\frac{v}{u}=\frac{v}{f}$

or $m=-\left(\frac{1}{f}\right)v+1$

comparing with y = mx + e

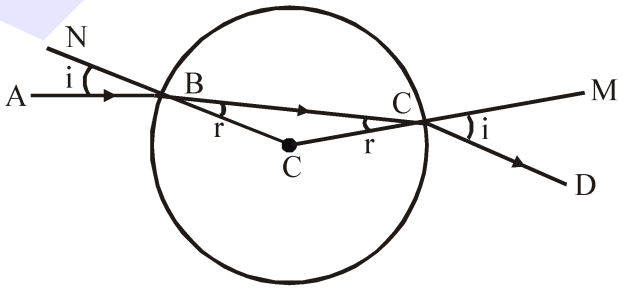
A,B & C are correct



11. Ans. (A,C)

12. Ans. (B,C)

Sol. The diagram shows the path of a ray *AB* incident at *B* on the sphere at angle *i*. The path *ABCD* shows the refracted ray *BC*, inside the sphere with angle *r* of refraction and the emergent ray *CD* with angle of emergence *i*.



Obviously $\angle OBC = \angle OCB = r$. Hence $\angle ABN = \angle$ DCM = i for all values of i.

Also, if $\angle ABN = 90°$, grazing incidence at B, then $\angle DCM = 90°$, grazing emergence at C.

But AB and CD will not be parallel.

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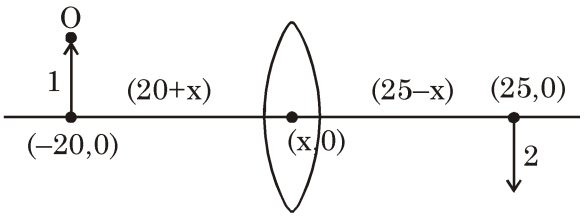
13. Ans. (A,B,C)

Sol. Light will emerge from prism if $A < \theta_C$

Light will not emerge from prism if $A > 2\theta_C$ $\therefore$ $\theta_C = 45°$

14. Ans. (B,C)

Sol.



Let's say the lens is located at (x, 0), then following is the representations of object and image. Assuming $x > 0$.

$$\Rightarrow \frac{1}{(25-x)} + \frac{1}{(20+x)} = \frac{1}{f} \qquad \text{... (i)}$$

$$\Rightarrow \frac{25-x}{20+x} = \frac{2}{1} = 2 \qquad \text{... (ii)}$$

By solving above two equations we get the answer.

15. Ans. (A,C,D)

Sol. The cavity will behave like a thick parallel slab and we know that it does not bring any deviations to light, it just laterally displaces the light and that does not matter for our case.

So independent of what is filled in the cavity the minimum angle of deviation will not change.

$$\Rightarrow \text{We know that, } u = \frac{\sin\left(\frac{A+\delta_{min}}{2}\right)}{\sin(A/2)}$$

$$\Rightarrow \sqrt{2} = \frac{\sin\left(\frac{60+\delta_{min}}{2}\right)}{\sin\left(\frac{60}{2}\right)}$$

$\Rightarrow \delta_{min} = 30°$ So, (A,C,D)

SECTION-III

Numerical Grid Type (Ranging from 0 to 9) 4 Q. [4 M (0)]

16. Ans. 6

Sol. Image of O will be formed at O itself.

Image of A will form, $\frac{1}{v} - \frac{3/2}{-5} = \frac{1-3/2}{-10} \Rightarrow \frac{1}{v} = \frac{1}{20} - \frac{3}{10} = \frac{-5}{20}$ $\therefore$ $v = -4$

$\therefore$ Distance between A & O = (10–4) = 6 cm

17. Ans. 4

Sol. $\frac{1}{F_{lens}} = (1.5-1)\left[\frac{1}{-40} - \frac{1}{-20}\right] = \frac{1}{80}$; $F_\ell = 80$ cm

$F_m = -\frac{20}{2} = -10$ cm; $\frac{1}{F_{eq}} = \frac{1}{f_m} - \frac{2}{f_\ell} = \frac{1}{-10} - \frac{2}{80}$; $f_{eq} = -8$ cm

Hence object should be placed at x = 16 cm, i.e. at the centre of curvature.

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18. Ans. 2

Sol. Refraction from a spherical surface is given by,

$$\frac{\mu_2}{v}=\frac{\mu_1}{u}=\frac{\mu_2-\mu_1}{R}$$

For refraction from the first surface,

$$\frac{7}{4v_1}-\frac{1}{-24}=\frac{\frac{3}{4}}{6}$$

For refraction from second surface,

$$\frac{4}{3v_2}-\frac{7}{4v_1}=\frac{\frac{3}{4}-\frac{7}{4}}{\infty}$$

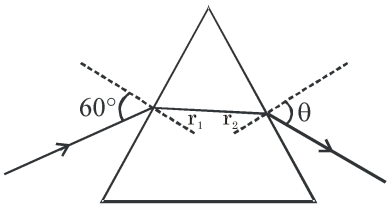
Adding the two equations gives,

$v_2 = 16$ cm

Thus distance from the bottom = depth – v_2 = 18 cm – 16 cm = 2 cm

19. Ans. 2

Sol.



From the geometry we have $r_1 + r_2 = 60°$

Using Snells law at the left side we have $\frac{\sin 60°}{\sin r_1}=n$

or

$$\sin r_1=\frac{\sqrt{3}}{2n}$$

Using Snells law at the right end

$$\frac{\sin r_2}{\sin\theta}=\frac{1}{n}$$

or

$\sin\theta = n \sin r_2$

$= n\sin(60° - r_1)$

$$=n\left(\frac{\sqrt{3}}{2}\cos r_1-\frac{1}{2}\sin r_1\right)$$

or

$$=n\left(\frac{\sqrt{3}}{2}\sqrt{1-\sin^2 r_1}-\frac{1}{2}\sin r_1\right)$$

or

$$=n\left(\frac{\sqrt{3}}{2}\sqrt{1-\frac{3}{4n^2}}-\frac{1}{2}\frac{\sqrt{3}}{2n}\right)$$

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Solving we get

$$\sin\theta = \frac{\sqrt{3}}{4}\sqrt{4n^2-3} - \frac{\sqrt{3}}{4}$$

Differentiating

$$\cos\theta d\theta = \frac{\sqrt{3}}{4} \times \frac{1}{2\sqrt{4n^2-3}} \times 8ndn$$

or

$$\frac{d\theta}{dn} = \frac{\sqrt{3}n}{\sqrt{4n^2-3}} \times \frac{1}{\cos\theta}$$

$$= \frac{\sqrt{3}\times\sqrt{3}}{\sqrt{4\times3-3}} \times \frac{1}{1/2}$$

$= 2$

SECTION-IV

Matrix Match Type (4 × 5) **3 Q. [8 M (for each entry +2(0)]**

20. Ans. (A) → (Q,S); (B) → (P,S); (C) → (Q,S); (D) → (S)

Sol. At minimum deviation $i_1 = i_2$, FG||BC
At maximum deviation $i_1 = 90^\circ$ or $i_2 = 90^\circ$
For $i_1 = 0$, TIR will not take place at AC

21. Ans. (A) S (B) P (C) R (D) Q

Sol. For (A) : $\frac{1}{f} = (\mu-1)\left(\frac{1}{p_1} = \frac{1}{p_2}\right) = (1.5-1)\left(\frac{1}{20}+\frac{1}{20}\right) = \frac{1}{20} \Rightarrow f = 20cm$

For (B) : $\frac{1}{f} = \left(\frac{1.5}{4/3}-1\right)\left(\frac{1}{20}+\frac{1}{20}\right) = \frac{1}{80} \Rightarrow f = 80\ cm$

For (C) : $\frac{1.5}{v_1} - \frac{4/3}{\infty} = \left(1.5-\frac{4}{3}\right)\left(\frac{1}{20}\right)$ & $\frac{1}{f} - \frac{1.5}{v_1}(1-1.5)\left(\frac{1}{-20}\right) \Rightarrow f = 30\ cm$

For (D) $\frac{1.5}{v_1} - \frac{1}{\infty} = (1.5-1)\left(\frac{1}{20}\right)$ & $\frac{4/3}{f} - \frac{1.5}{v_1} = \left(\frac{4}{3}-1.5\right)\left(-\frac{1}{20}\right) \Rightarrow f = 40cm$

22. Ans. (A) PRST, (B) PQ; (C) S; (D) PQ

Subjective Type **6 Q. [4 M (0)]**

23. Ans. 60°, 30°, 2, $\frac{4}{\sqrt{3}}$

24. Ans. (n - 1)R/(3n - 1)

25. Ans. [Number of images = 2, distance between them = 0.003 m]

26. Ans. $y_1 = f\,\delta_1 = 20 \times \frac{\pi}{100}$ cm = π/5 cm ; $y_2 = f\,\delta_2 = 20 \times \frac{3\pi}{50}$ cm = 1.2π cm

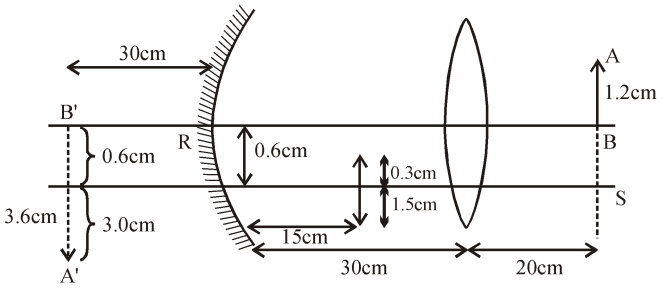
27. Ans. 20 cm behind the mirror

Sol. Parallel ray on equi-concave lens will make the image on the left side of the lens at it's focal length. This will be considered an object for the plane mirror. This object will be at the distance of |d + f| from the plane mirror on the left so image will be |d + f| on the right side of mirror.

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28. **Ans.** 15 cm right of the mirror, magnification = 1.5; (–15 cm, –1.5 cm; – 15 cm, 0.3 cm)

Sol.



For lens,

$u = -20$ cm, $f = +15$ cm

$$\frac{1}{v}-\frac{1}{u}=\frac{1}{f} \Rightarrow v_1 = \frac{uf}{u+f}$$

$= 60$ cm left of lens

This A′B′ will act as object for mirror.

So, $u_2 = +30$ cm

$f_2 = -30$ cm

$$\frac{1}{v_2}+\frac{1}{u_2}=\frac{1}{f_2} \Rightarrow v_2 = \frac{u_2 f_2}{u_2 - f_2} = -15 \text{ cm right of mirror}$$

Position of A″ = (15 cm, –1.5 cm)

B″ = (15 cm, +0.3 cm)

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